HIGH-LEVEL DESIGN

Biotech Labs Exhibition & Inquiry Portal

Draft v0.1 | QR-led exhibition pilot with a reusable web inquiry platform

Field	Value
Document purpose	Describe the proposed logical architecture and delivery approach for the portal pilot.
Prepared for	Business and technology stakeholders
Status	Draft for review
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1. Executive summary

This High-Level Design (HLD) defines a responsive portal for collecting supplier and buyer inquiries at an exhibition through a QR code, then reusing the same journey for normal website inquiries. The portal presents a clear Sell to us or Purchase from us choice, captures structured information, and routes each submission to the appropriate internal process.

The design separates the visitor-facing portal from workflow, catalogue, data, administration, and enterprise-integration services. This keeps the exhibition pilot lightweight while allowing the same capability to scale into a controlled lead-generation and vendor-onboarding channel.

Design intent: Use one configurable platform for both routes. AI helps users complete information; it does not autonomously approve vendors, create records, or make commercial decisions.

2. Scope and objectives

Area	HLD position
In scope	QR-led exhibition entry, normal web inquiry entry, supplier intake, buyer inquiry intake, catalogue upload, location evidence with consent, admin review, vendor creation trigger, marketing lead routing, Excel export, and audit logging.
Planned AI assist	Multilingual guided voice input and visiting-card data extraction, with reviewable, editable fields before submission.
Proposed pilot exclusions	Online ordering, payment processing, automated vendor approval, commercial pricing workflows, and AI-led business decisions.

Business outcome: Convert exhibition interactions into structured supplier candidates and qualified purchase leads without creating a separate temporary system.

3. Functional design

Journey	High-level behaviour
Sell to us	Visitor selects one or more departments, chooses relevant product types, provides supplier details and consented location evidence, then uploads a digital catalogue. Images are standardized into a PDF where needed. The submission enters an admin review queue; only an approved record can be added to the enterprise vendor platform.
Purchase from us	Visitor selects listed products or enters an exact requirement, identifies a pharma category such as IP, USP, PP, or another configured category, provides buyer details and consented location evidence, and submits. The inquiry becomes a marketing lead and remains available for controlled Excel export.
Shared journey	Both routes use the same responsive portal, form validation, confirmation message, audit trail, optional AI assistance, and configurable business taxonomy.

3.1 Journey control points

The portal must save the selected route early, show only relevant fields, validate required information before submission, and issue a reference confirmation. Configured business taxonomies determine which departments, product types, categories, and downstream rules apply.

3.2 Visitor assistance and fallback

Voice input and visiting-card extraction are optional accelerators. A user must be able to review or correct extracted fields and complete the form manually if camera, microphone, network, or location access is unavailable or declined.

3.3 File and evidence handling

Catalogue files are accepted through a controlled upload path. If a visitor supplies images, the portal preserves approved originals and creates a standard PDF package for review. Location evidence is captured only after consent and is stored with the submission timestamp.

4. Logical system architecture

The architecture below groups the solution by channel, portal, workflow services, secure data layer, internal administration, and enterprise integrations. It intentionally avoids selecting a final cloud provider, CRM, ERP, or AI vendor; those are implementation decisions to confirm during detailed design.

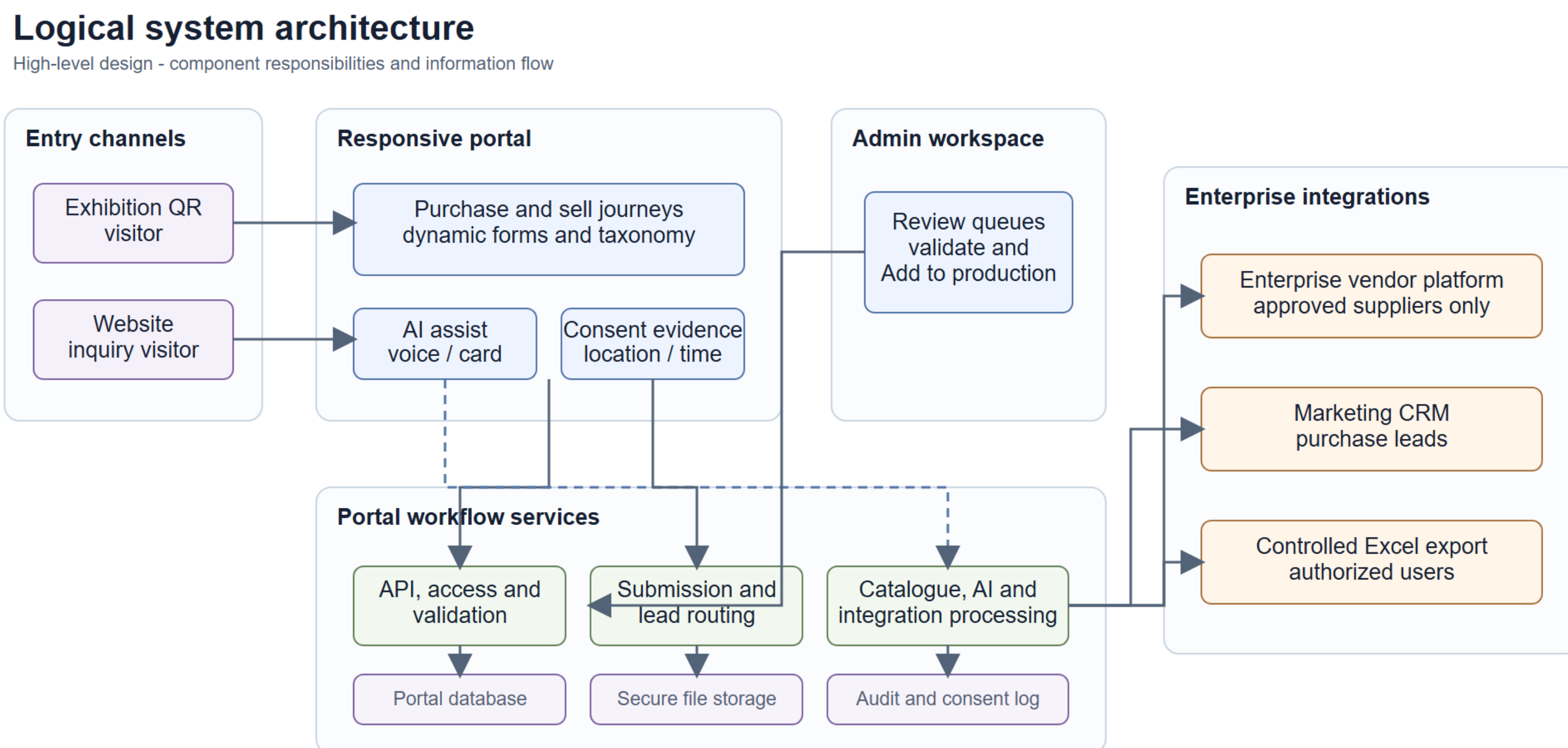

Figure 1. Proposed logical system architecture

5. Component responsibilities

Component	Responsibility
Responsive web portal	Presents the purchase and sell journeys; renders configurable departments, product types, pharma categories, consent capture, and file-upload controls.
Portal API and access control	Accepts portal and admin requests, applies authentication and authorization, validates input, and protects endpoints from misuse.
Workflow and routing	Creates a submission, runs validation and deduplication checks, assigns a supplier-review or purchase-lead path, and maintains status.
Catalogue processing	Receives approved file types, performs security checks, stores files securely, and converts image bundles to a standard PDF package when required.
AI orchestration	Supports voice-to-form and visiting-card extraction. It returns suggested field values with confidence or review indicators; it must not silently overwrite user-confirmed data.
Admin workspace	Lets internal users search queues, review supporting files, resolve duplicate records, and explicitly authorize Add to production for suppliers.
Integration service	Creates approved vendor records in the enterprise platform, dispatches purchase leads to marketing/CRM, and produces governed Excel exports.

5.1 Ownership boundary

The portal remains the system of record for the original inquiry, consent/evidence, approval history, and routing outcome. The enterprise vendor platform remains the system of record for approved vendor master data; the CRM or lead destination remains the system of record for commercial follow-up.

6. Information and workflow design

6.1 Records and storage

Record group	Minimum high-level contents
Submission profile	Route selected, contact and company details, selected departments/products/categories, workflow status, and internal owner.
Location evidence	Consent flag, timestamp, and permitted evidence source such as browser location or network-derived evidence. Precision, retention, and fallback rules remain policy decisions.
Catalogue assets	Original approved uploads, derived PDF package where applicable, processing status, checksum, and access-control metadata.
Audit events	Submission, consent, edits, review decisions, vendor-creation attempts, lead dispatch, export generation, and integration outcomes.

6.2 Supplier workflow

A supplier submission is stored as a candidate record. The admin workspace validates information, identifies duplicates, reviews the catalogue, and decides whether to Add to production. The integration service creates or updates the enterprise vendor record only after that explicit approval. Failures are retained for retry and visible review; a source submission is never silently discarded.

6.3 Purchase-lead workflow

A purchase inquiry is stored as a lead and sent to the configured marketing destination either immediately or by a scheduled batch. The original record is retained even if delivery fails. Authorized users can export a controlled Excel report that reflects the same governed data set.

7. Integration design

Integration	Trigger	Direction	Control
Enterprise vendor platform	Admin-approved supplier	Outbound create or update	Idempotent request key, approval state, response/audit capture, retry queue.
Marketing CRM or lead inbox	Submitted purchase inquiry	Outbound lead dispatch	Configured routing rule, duplicate strategy, delivery status, retry queue.
Excel reporting	Authorized admin request or schedule	Generated export	Role-based access, row-level audit event, controlled file retention.
AI provider	Optional voice or card scan	Request/response assist	User consent, minimal data sent, response review, retention policy, provider assessment.

8. Security, privacy and operational controls

Access control: Separate public visitor access from authenticated admin access. Apply role-based permissions for review, approval, export, and configuration activities.

Consent and privacy: Obtain a clear affirmative consent for location evidence and visiting-card extraction. Store only permitted evidence, state the purpose, and use a configured retention policy.

File security: Allowlist file types and sizes, scan uploads, store files outside the public web tier, and issue time-limited access only to authorized users.

Auditability: Record submission, review, consent, export, and integration events so that an exhibition lead can be traced from source to business action.

Resilience: Use retryable, observable integration jobs. Capture a clear status for vendor creation and lead delivery so operational teams can resolve exceptions.

9. Non-functional requirements and design guardrails

Area	HLD requirement
Usability	Mobile-first and fast to complete in an exhibition setting; manual form completion remains available if voice, camera, or location access is declined.
Configurability	Business users should be able to manage department, product type, and pharma-category taxonomies without redeploying the portal.
Scalability	Separate file processing and outbound integrations from visitor request handling so an exhibition surge does not block submissions.
Observability	Measure completion, abandonment, upload failures, AI-assist usage, queue age, integration failures, and export activity.
Availability	Pilot needs, fallback behaviour, support ownership, and formal service targets must be agreed before production rollout.

10. Delivery approach

Phase	Primary deliverables
Phase 1 - Exhibition pilot	QR entry, responsive forms, file upload, consented evidence capture, admin review queues, controlled Excel export, and manual/limited integration handling.
Phase 2 - Operational integration	Harden enterprise vendor creation, CRM routing, scheduled lead distribution, monitoring, configuration management, and audit reporting.
Phase 3 - Full inquiry portal	Expose the same journeys through normal website entry points, refine taxonomies and conversion flows using pilot data, and activate the selected AI assistance features.

11. Assumptions, dependencies and decisions required

Decision or dependency	What must be agreed
Enterprise vendor interface	Confirm available API, identity matching rules, approval ownership, and whether create versus update is permitted.
Marketing destination	Confirm CRM or mailbox destination, mandatory lead fields, routing owner, delivery frequency, and follow-up SLA.
Taxonomy ownership	Confirm the final department list, department-to-product mapping, and pharma categories. Treat this as business-managed configuration.
Location-evidence policy	Confirm lawful purpose, consent wording, allowed sources, precision, fallback behaviour, retention period, and deletion process.
AI provider and data policy	Confirm supported languages, card/image handling, accuracy expectations, human review process, and acceptable data-processing terms.
Exhibition connectivity	Confirm expected network conditions, operator support plan, QR placement, fallback instructions, and device/browser coverage.

Recommended next step: Hold a cross-functional HLD review to confirm the enterprise interfaces, taxonomy owner, consent policy, AI data policy, exhibition connectivity plan, and pilot support model before Low-Level Design begins.

Appendix A. User and operations flow

This workflow is included as a business-flow reference. Figure 1 remains the authoritative HLD architecture view; the appendix clarifies how individual portal submissions become supplier candidates or marketing leads.

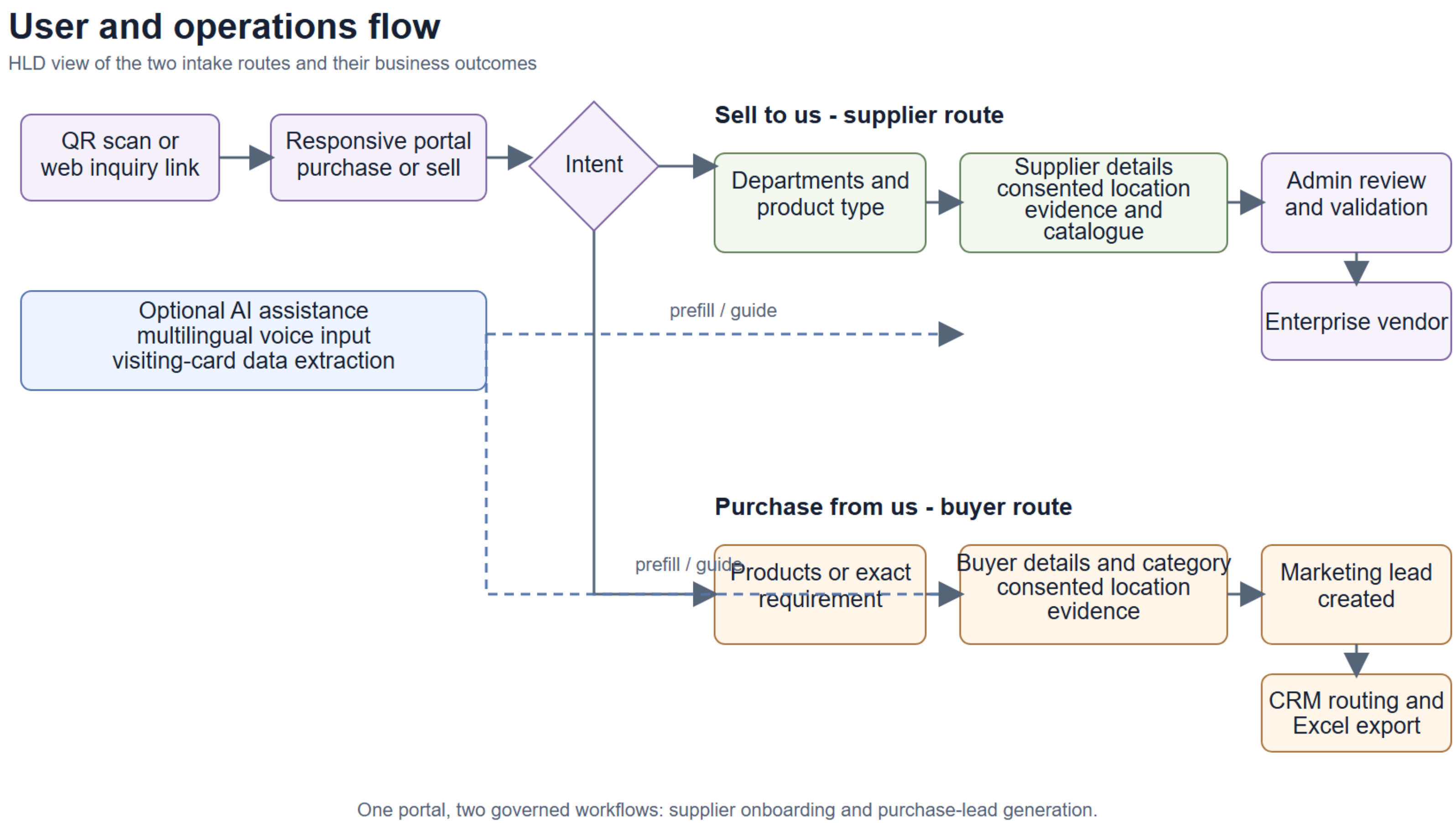

Figure 2. Tentative user and operations flow